

Problem Statement

Client A is a multi-brand e-commerce group operating four consumer product brands across separate Shopify stores, with paid media running on both Google Ads and Facebook Ads. Despite significant advertising investment, leadership had no unified view of performance across brands or channels. Without a consolidated **data model** linking spend, traffic, and revenue, two critical problems prevented informed decision-making:

Fragmented Data Ecosystem

Four brands, three ad platforms, and multiple Shopify stores operated in complete isolation – no cross-brand reporting, no unified KPIs.

No Customer Lifetime Visibility

No LTV modelling, no new vs. returning customer segmentation, and no cohort tracking to validate acquisition economics or justify media spend levels.

Solution & Objectives

The solution was to design and build a **7-stage ELT pipeline** in Google BigQuery, unifying all data sources into a single analytics layer and delivering four purpose-built Looker Studio dashboards for different stakeholder needs.

Primary	Unify all brand data into a single, repeatable BigQuery model with automated daily refresh.
Efficiency	Quantify paid media ROI by channel, brand, and customer type (new vs. returning).
LTV	Model customer lifetime value at 1, 2, 3, 6 and 12-month horizons per acquisition cohort.
Insight	Deliver executive, media, and cohort dashboards enabling data-driven budget decisions.

Pipeline Workflow — 7-Stage ELT

The pipeline follows a sequential Extract-Load-Transform architecture, with each stage materialised as a BigQuery table to avoid recomputation and ensure traceability.

1	sales_performance	Order-level revenue, COGS & LTV model base
2	traffic_performance	GA4 traffic unified across all 4 brands
3	media_performance	Spend cleaned & consolidated from 7 ad tables
4	performance_merged_full	Traffic + media joined via FULL OUTER
5	daily_brand_metrics	Brand-level P&L with net profit & NCROAS
6	daily_paid_performance	Channel-level blended MER & acquisition metrics
7	funnel_long	UNPIVOTed funnel for Looker Studio visualisation

Data Sources & Input Categories

Shopify	Order data, customer status, SKU-level COGS lookup
Google Analytics 4	Sessions, users, bounce rate, add-to-cart, channel source
Google Ads	Impressions, clicks, spend per brand (4 accounts)
Facebook Ads	Impressions, clicks, spend per brand (3 accounts)

Key Outcomes

→ Single Source of Truth

All four brands report from one unified BigQuery dataset, eliminating manual reconciliation.

→ Profitability Visibility

Gross and net profit calculated daily per brand and channel, including COGS deduction.

→ New vs. Returning Segmentation

NCROAS, NCCPA and NCAOV tracked separately, enabling precise acquisition optimisation.

→ LTV-Driven Strategy

Multi-horizon LTV curves (1M–12M) allow teams to set maximum sustainable CAC thresholds.

→ Automated Reporting

Scheduled queries replace days of manual work with a fully repeatable pipeline.

Dashboard Suite

1. Executive Overview
- KPI scorecards, brand comparison table, conversion funnels (Impressions → Purchases).
2. Paid Performance
- Channel matrix (ROAS, NCCPA by brand/platform) and MoM spend vs. ROI trends.
3. CAC Toleration & LTV
- CAC pay-back model, net margin tracking, LTV_1M – LTV_12M curves.
4. Cohort Analysis
- Acquisition cohort heat map and cumulative LTV trend lines by cohort month.

Skills & Tools

Google BigQuery	SQL / ELT
Looker Studio	GA4
Google Ads API	Meta Ads
Shopify Data	Python
LTV Modelling	
Cohort Analysis	
Data Engineering	English

About This Project

Client	Anonymous (Client A)
Industry	Multi-Brand E-Commerce
Pipeline	7-Stage BigQuery ELT
KPI Proxy	Revenue, LTV, NCROAS
Dashboards	4 Looker Studio views
Year	2024–2025

Confidentiality

The client’s identity has been anonymised as “Client A” in accordance with a confidentiality agreement. Brand names, GCP project identifiers, account IDs, and all financial data remain proprietary and are not disclosed.